5–Data-Characterization

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2026-07-24

Setup

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(hrbrthemes) #point density charts
```

```
## Warning: package 'hrbrthemes' was built under R version 4.5.3
```

```r
library(viridis)
```

```
## Warning: package 'viridis' was built under R version 4.5.2
```

```
## Loading required package: viridisLite
```

```r
library(sf)
```

```
## Linking to GEOS 3.13.1, GDAL 3.11.0, PROJ 9.6.0; sf_use_s2() is TRUE
```

```r
library(here)
```

```
## Warning: package 'here' was built under R version 4.5.2
```

```
## here() starts at C:/Users/Elizabeth Buhr/OneDrive/Documents/fuels-analysis
```

```r
transects <- st_read(here("data/processed/fuels_analysis.gpkg"), layer = "transectsDS")
```

```
## Reading layer `transectsDS' from data source
##   `C:\Users\Elizabeth Buhr\OneDrive\Documents\fuels-analysis\data\processed\fuels_analysis.gpkg'
##   using driver `GPKG'
## Simple feature collection with 3934 features and 27 fields
## Geometry type: POINT
## Dimension:     XY
## Bounding box:  xmin: 173613.9 ymin: 4094767 xmax: 555237.8 ymax: 4512971
## Projected CRS: NAD83 / UTM zone 13N
```

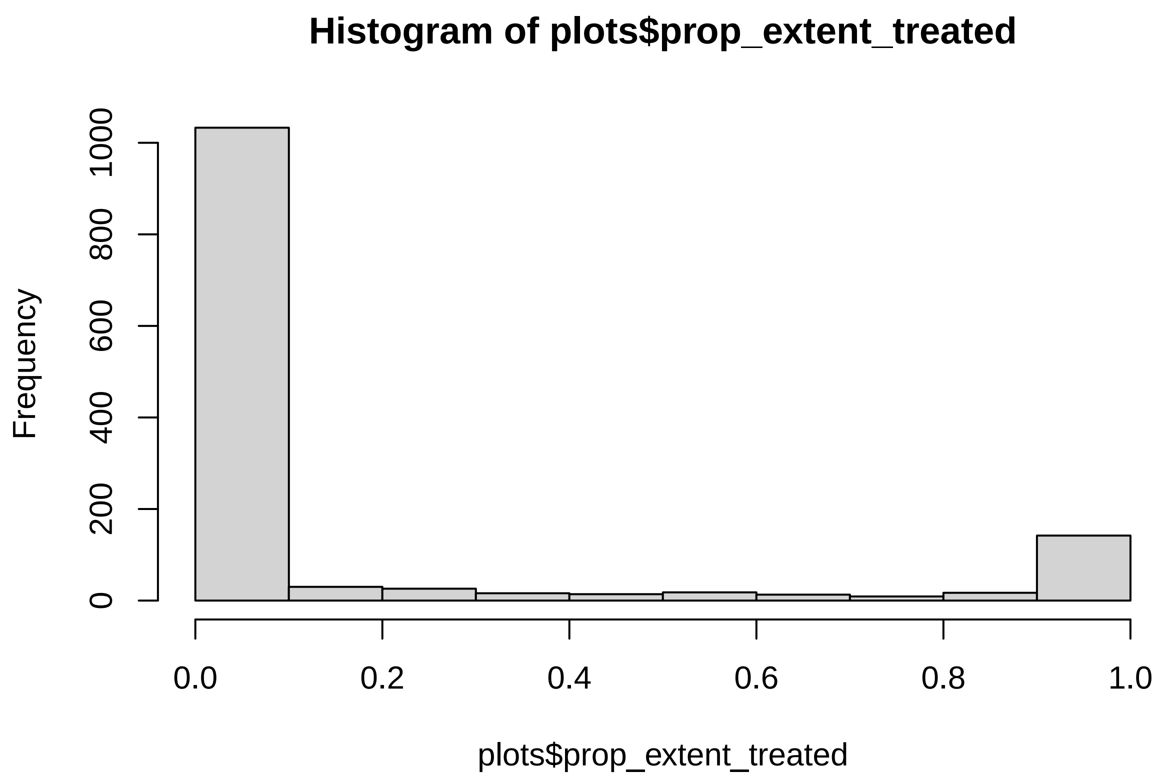
```r
transects$TCC <- as.numeric(transects$TCC)
transects$TRT_TYPE[is.na(transects$TRT_TYPE)] <- "Untreated"

plots <- st_read(here("data/processed/fuels_analysis.gpkg"), layer = "allplotsDS")
```

```
## Reading layer `allplotsDS' from data source
##   `C:\Users\Elizabeth Buhr\OneDrive\Documents\fuels-analysis\data\processed\fuels_analysis.gpkg'
##   using driver `GPKG'
## Simple feature collection with 1318 features and 37 fields
## Geometry type: POLYGON
## Dimension:     XY
## Bounding box:  xmin: 172907.8 ymin: 4095065 xmax: 555333.6 ymax: 4513480
## Projected CRS: NAD83 / UTM zone 13N
```

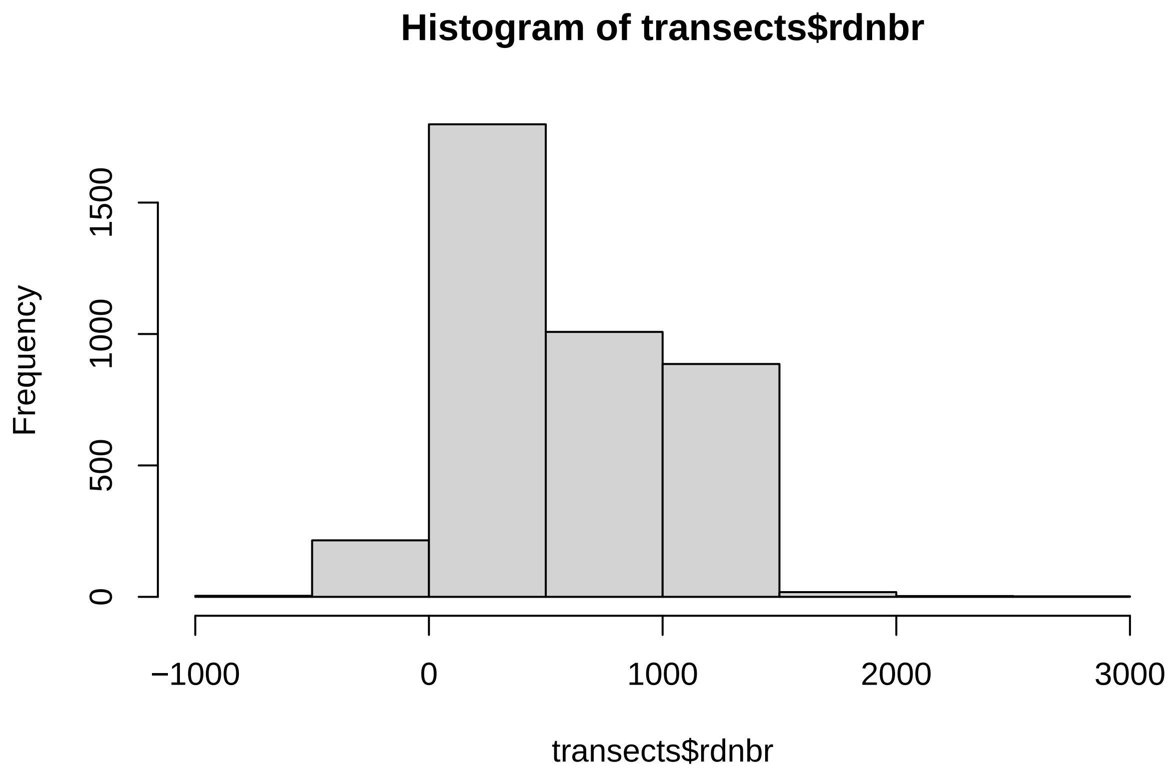
Summary Table of Treatment Type and Mean rdnbr for Transects and Plots

```r
# # Summary for transects
# transect_summary <- transects %>%
#   st_drop_geometry() %>%
#   group_by(TRT_TYPE) %>%
#   summarise(
#     n = n(),
#     mean_rdnbr = round(mean(rdnbr, na.rm = TRUE), 3),
#     sd_rdnbr = round(sd(rdnbr, na.rm = TRUE), 3),
#     min_rdnbr = round(min(rdnbr, na.rm = TRUE), 3),
#     max_rdnbr = round(max(rdnbr, na.rm = TRUE), 3)
#   ) %>%
#   mutate(source = "Transects")
#
# # Summary for plots
# plot_summary <- plots %>%
#   st_drop_geometry() %>%
#   group_by(prop_extent_treated) %>%
#   summarise(
#     n = n(),
#     mean_rdnbr = round(mean(mean_rdnbr, na.rm = TRUE), 3),
#     sd_rdnbr = round(sd(mean_rdnbr, na.rm = TRUE), 3),
#     min_rdnbr = round(min(mean_rdnbr, na.rm = TRUE), 3),
#     max_rdnbr = round(max(mean_rdnbr, na.rm = TRUE), 3)
```

```r
#   ) %>%
#   mutate(source = "Plots")
#
# # Combined table
# bind_rows(transect_summary, plot_summary) %>%
#   select(source, TRT_TYPE, n, mean_rdnbr, sd_rdnbr, min_rdnbr, max_rdnbr) %>%
#   arrange(source, TRT_TYPE) %>%
#   knitr::kable(
#     col.names = c("Source", "Treatment Type", "N", "Mean rdnbr", "SD rdnbr", "Min rdnbr", "Max rdnbr"
#     caption = "Summary of Treatment Type and rdnbr by Data Source"
#   )
#
hist(plots$prop_extent_treated)
```




```r
hist(transects$rdnbr)
```



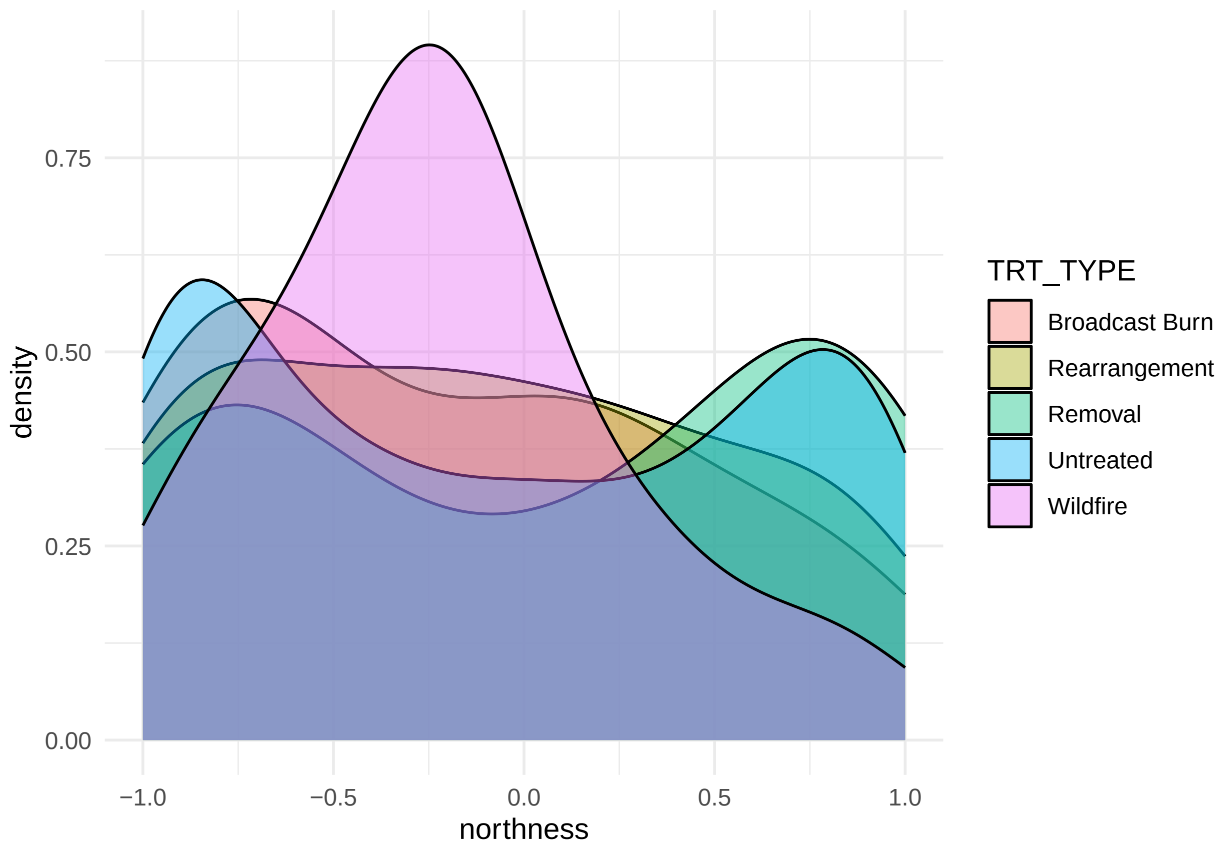

```r
range(transects$rdnbr)
```

```
## [1] -767 2943
```

Point Density of Transect Characteristics by Treatment Type

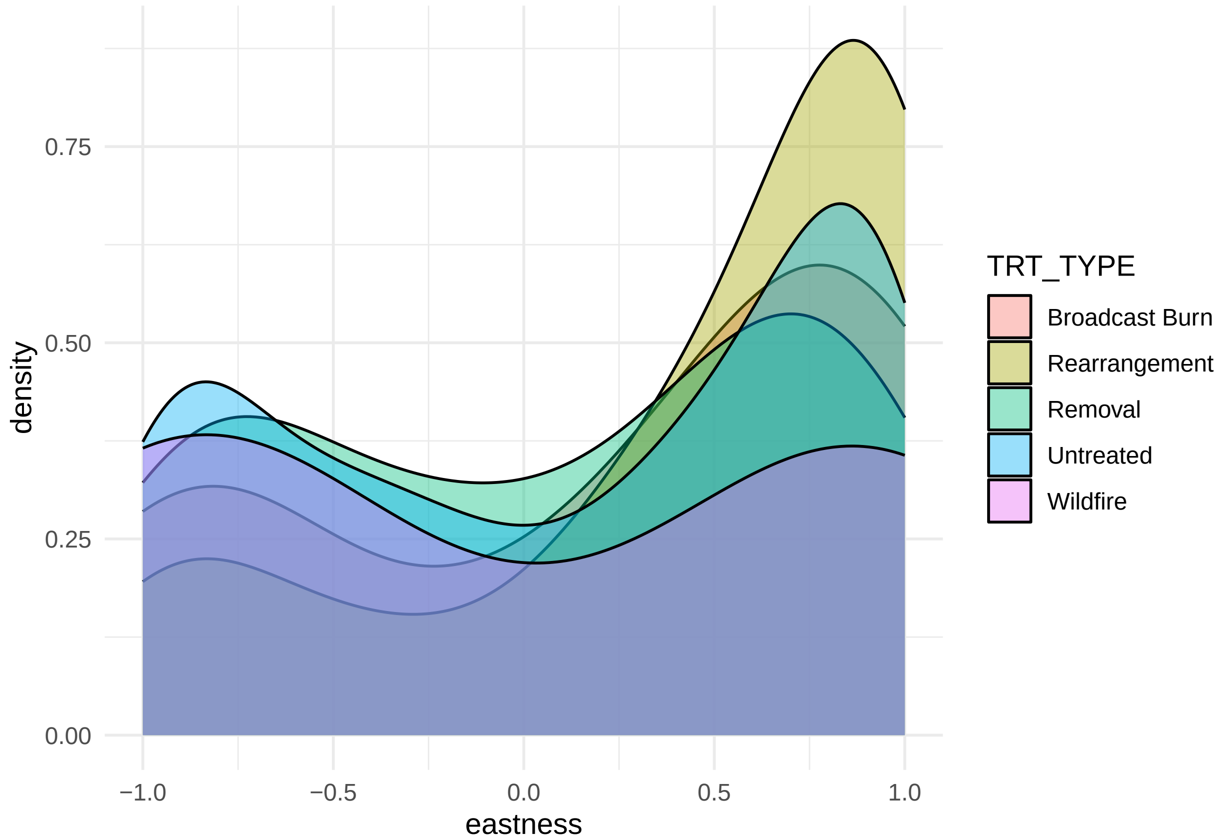
```r
#Topography
p1 <- ggplot(data=transects, aes(x=northness, group=TRT_TYPE, fill=TRT_TYPE)) +
    geom_density(adjust=1.5, alpha=.4) +
    theme_minimal()
p1
```

```
## Warning: Removed 149 rows containing non-finite outside the scale range
## (`stat_density()`).
```

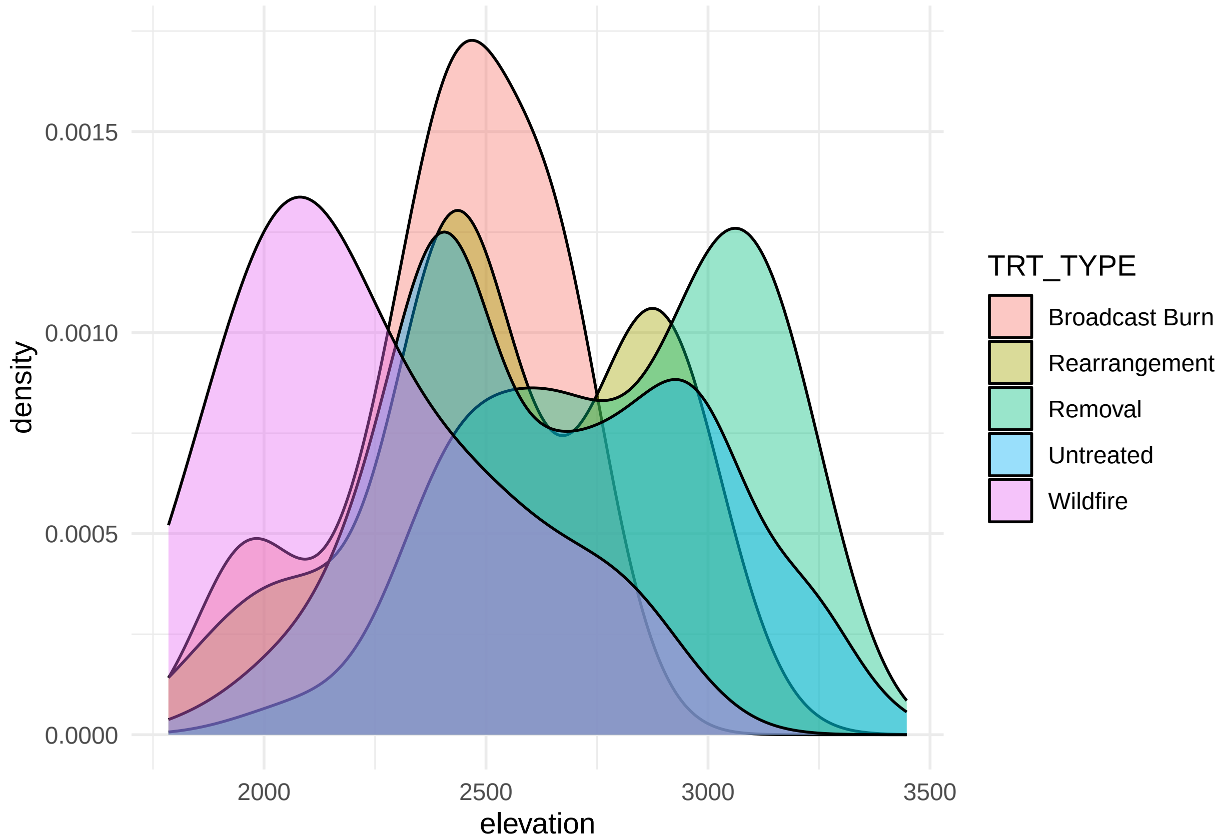



```
p2 <- ggplot(data=transects, aes(x=eastness, group=TRT_TYPE, fill=TRT_TYPE)) +
    geom_density(adjust=1.5, alpha=.4) +
    theme_minimal()
p2
```

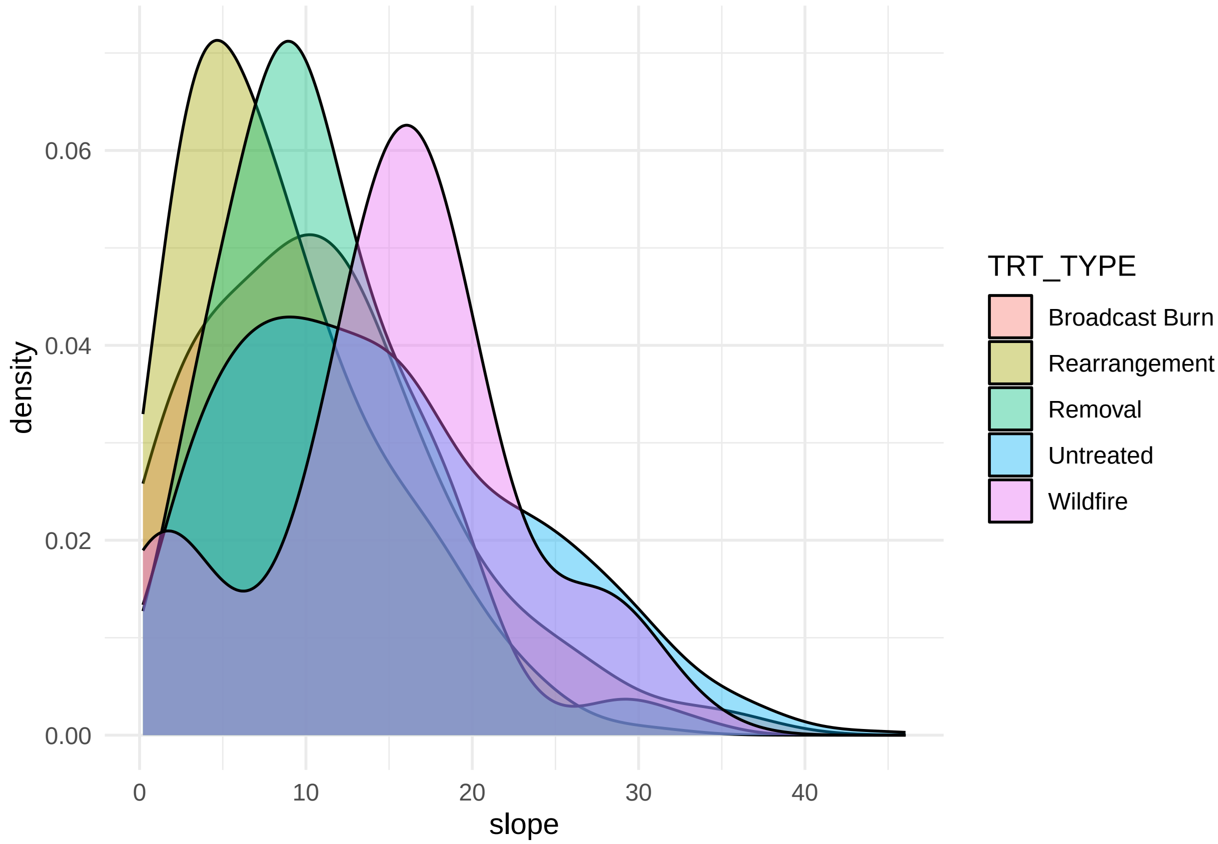
```
## Warning: Removed 149 rows containing non-finite outside the scale range
## (‘stat_density()‘).
```




```r
p3 <-  ggplot(data=transects, aes(x=elevation, group=TRT_TYPE, fill=TRT_TYPE)) +
    geom_density(adjust=1.5, alpha=.4) +
    theme_minimal()
p3
```

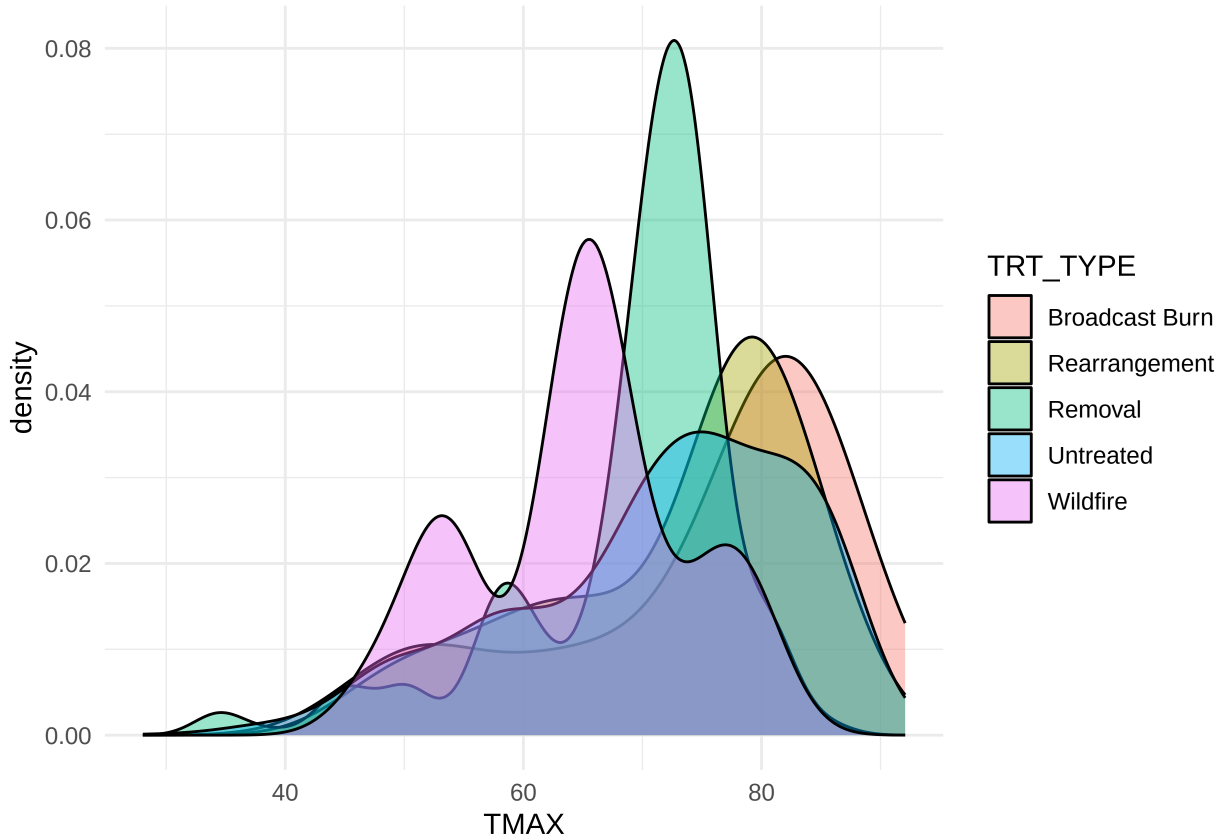



```
p4 <-  ggplot(data=transects, aes(x=slope, group=TRT_TYPE, fill=TRT_TYPE)) +
    geom_density(adjust=1.5, alpha=.4) +
    theme_minimal()
p4
```



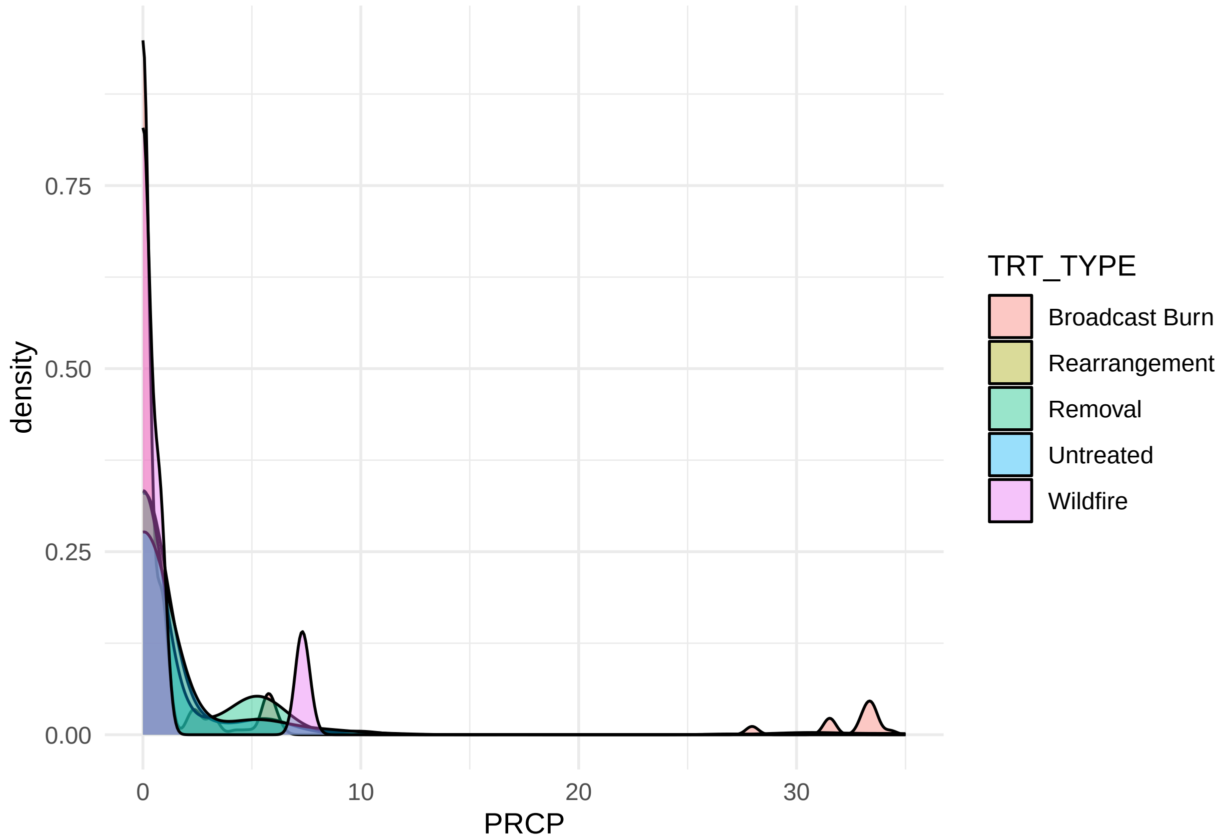

```
#Weather and Climate
p5 <- ggplot(data=transects, aes(x=TMAX, group=TRT_TYPE, fill=TRT_TYPE)) +
    geom_density(adjust=1.5, alpha=.4) +
    theme_minimal()
p5
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_density()`).
```

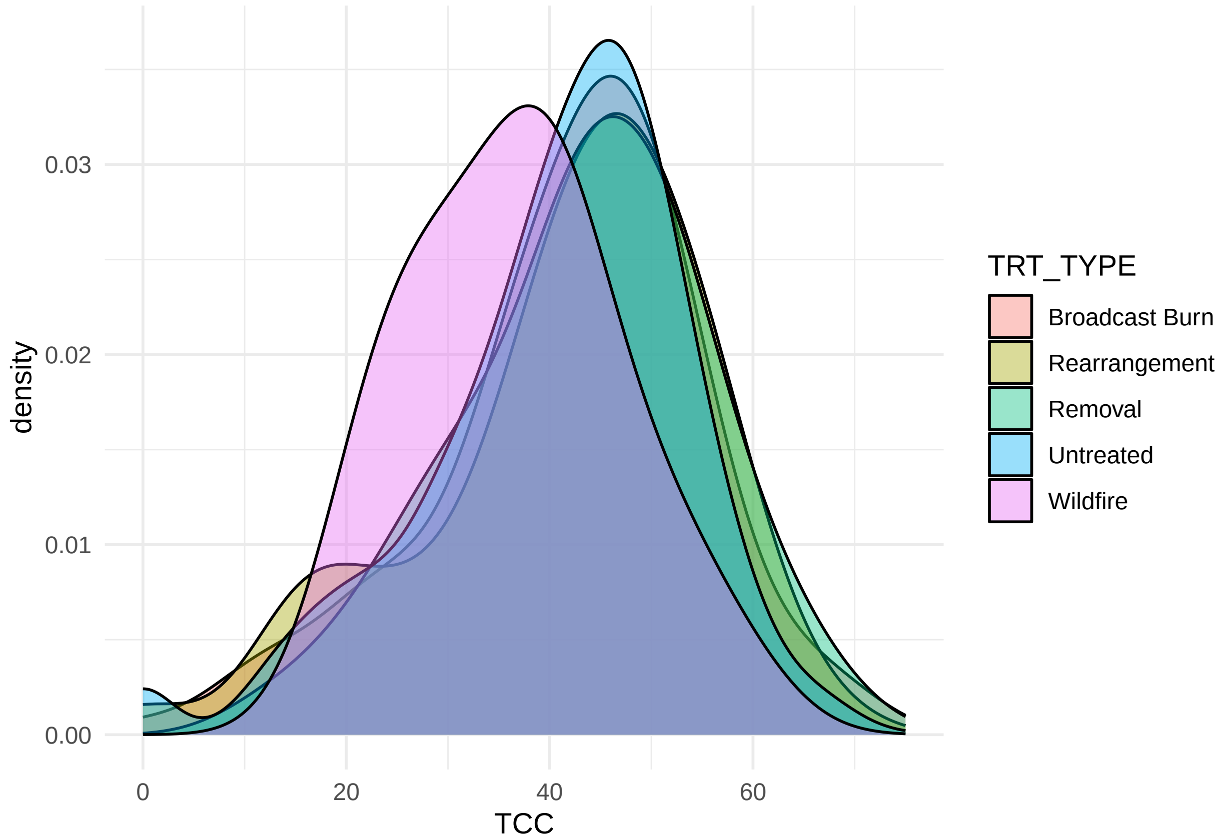



```
p6 <- ggplot(data=transects, aes(x=PRCP, group=TRT_TYPE, fill=TRT_TYPE)) +
    geom_density(adjust=1.5, alpha=.4) +
    theme_minimal()
p6
```

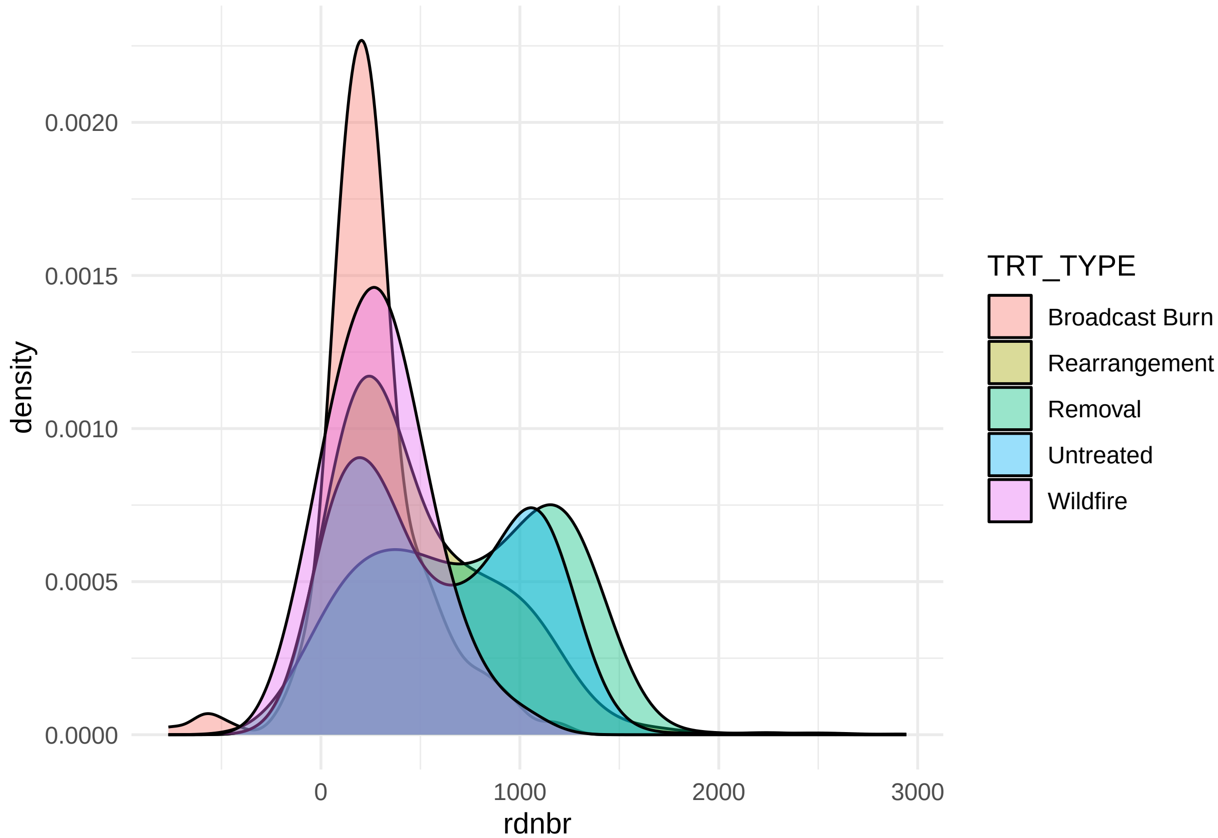
```
## Warning: Removed 1 row containing non-finite outside the scale range
## (‘stat_density()‘).
```



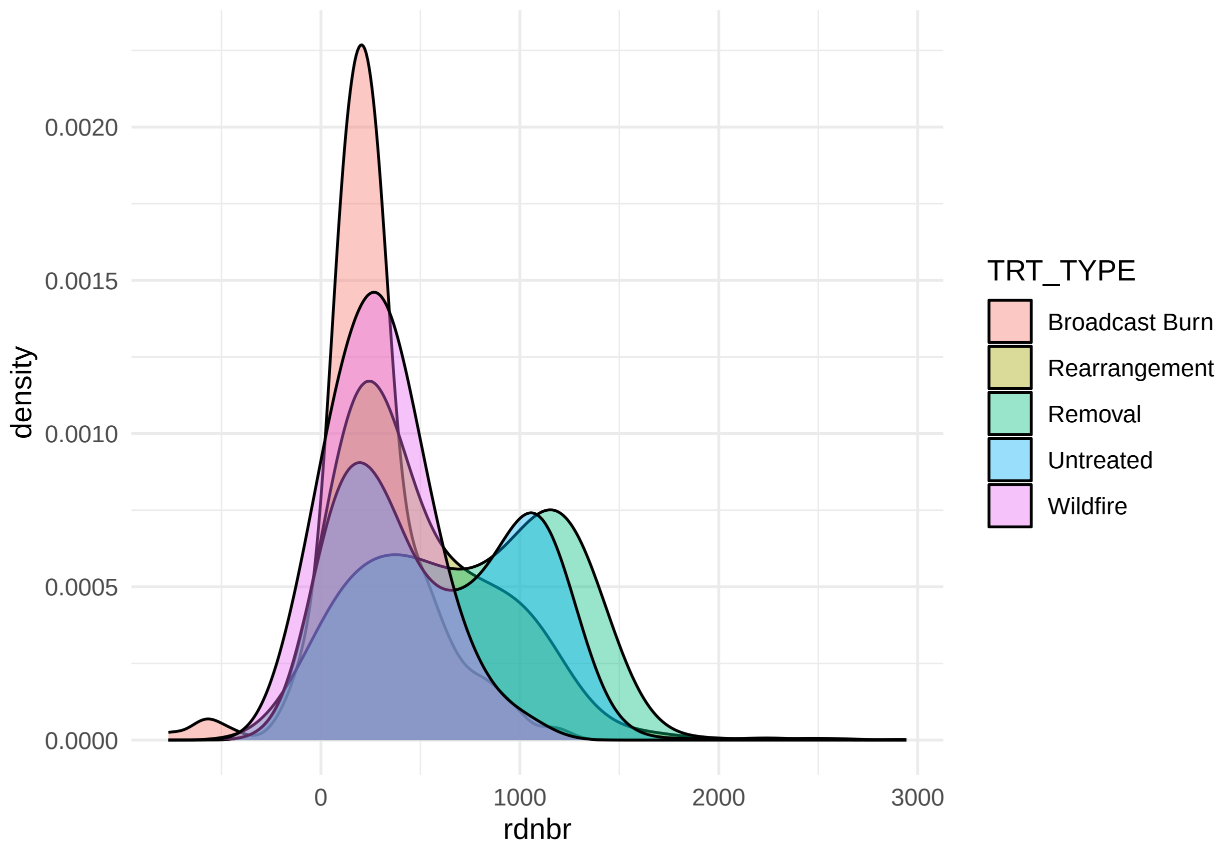

```
#Fuels
p8 <- ggplot(data=transects, aes(x=TCC, group=TRT_TYPE, fill=TRT_TYPE)) +
    geom_density(adjust=1.5, alpha=.4) +
    theme_minimal()
p8
```




```r
#Response
p12 <- ggplot(data=transects, aes(x=rdnbr, group=TRT_TYPE, fill=TRT_TYPE)) +
    geom_density(adjust=1.5, alpha=.4) +
    theme_minimal()
p12
```



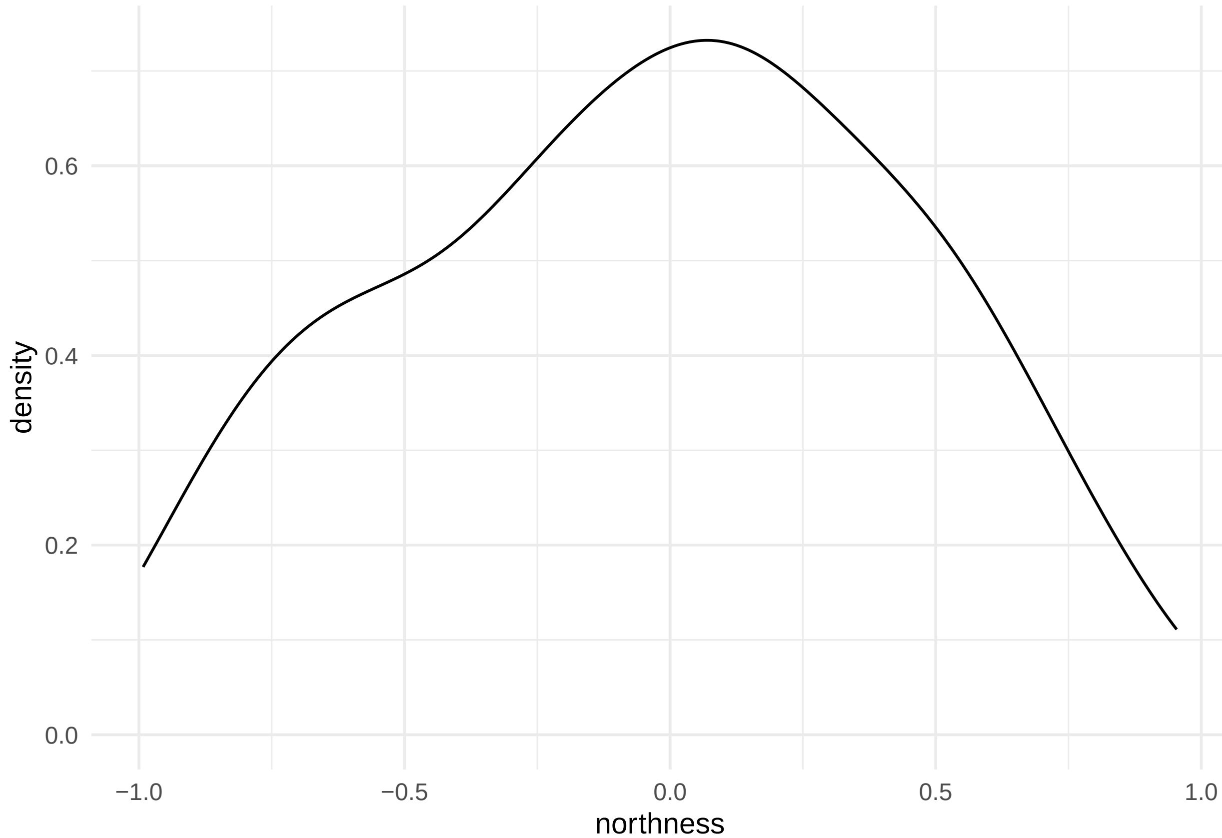

```r
p13 <- ggplot(data=transects, aes(x=rdnbr, group=TRT_TYPE, fill=TRT_TYPE)) +
    geom_density(adjust=1.5, alpha=.4) +
    theme_minimal()
p13
```



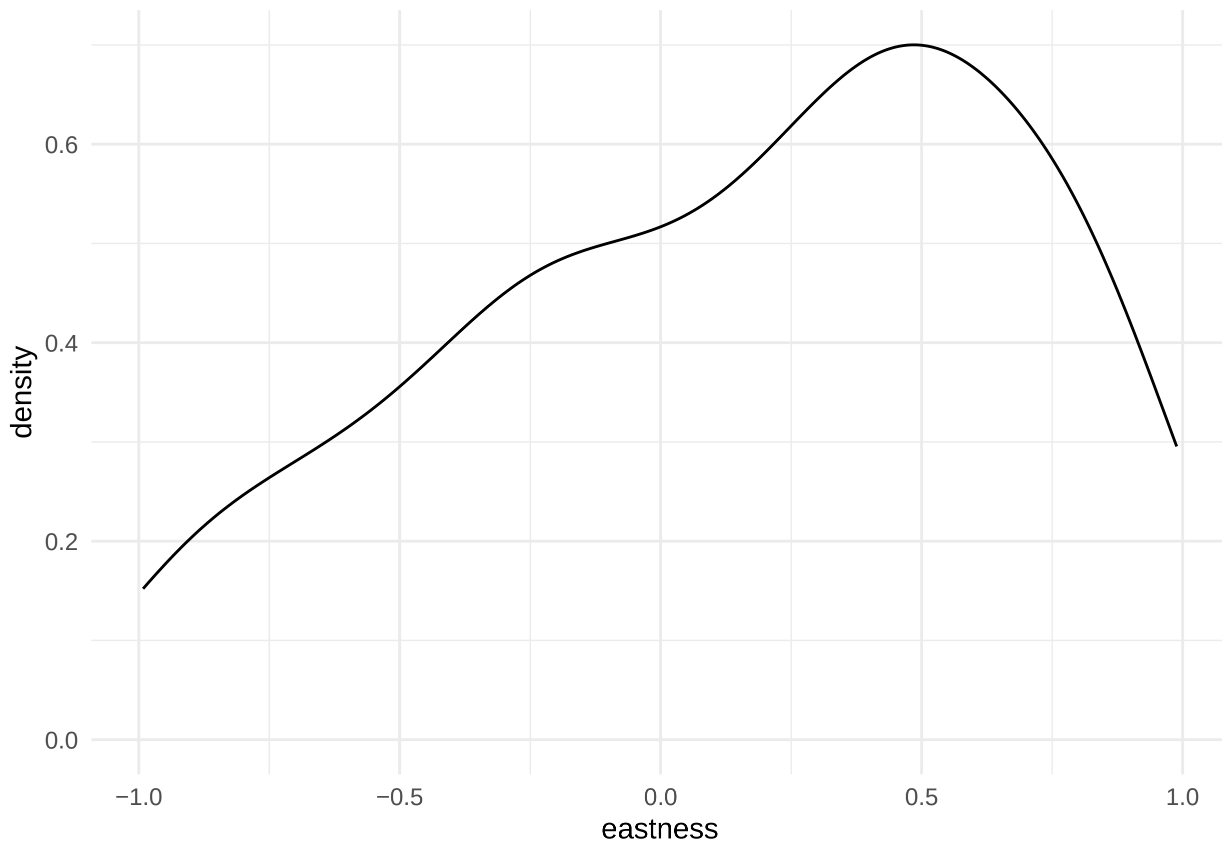
Point Density of Plot Characteristics by Treatment Type

```
## Point Density of Plot Characteristics by Treatment Type
vars_to_plot <- c("northness", "eastness", "elevation", "slope", "TMAX", "PRCP", "TOTAL_COUNT", "prop_Re
                  "prop_Wildfire", "prop_Broadcast.Burn", "prop_Removal",
                  "prop_extent_treated", "TCC", "mean_rdnbr")

plot_list <- map(vars_to_plot, ~{
  ggplot(data = plots, aes(x = .data[[.x]])) +
    geom_density(adjust = 1.5, alpha = .4) +
    labs(x = .x) +
    theme_minimal()
})

walk(plot_list, print)
```

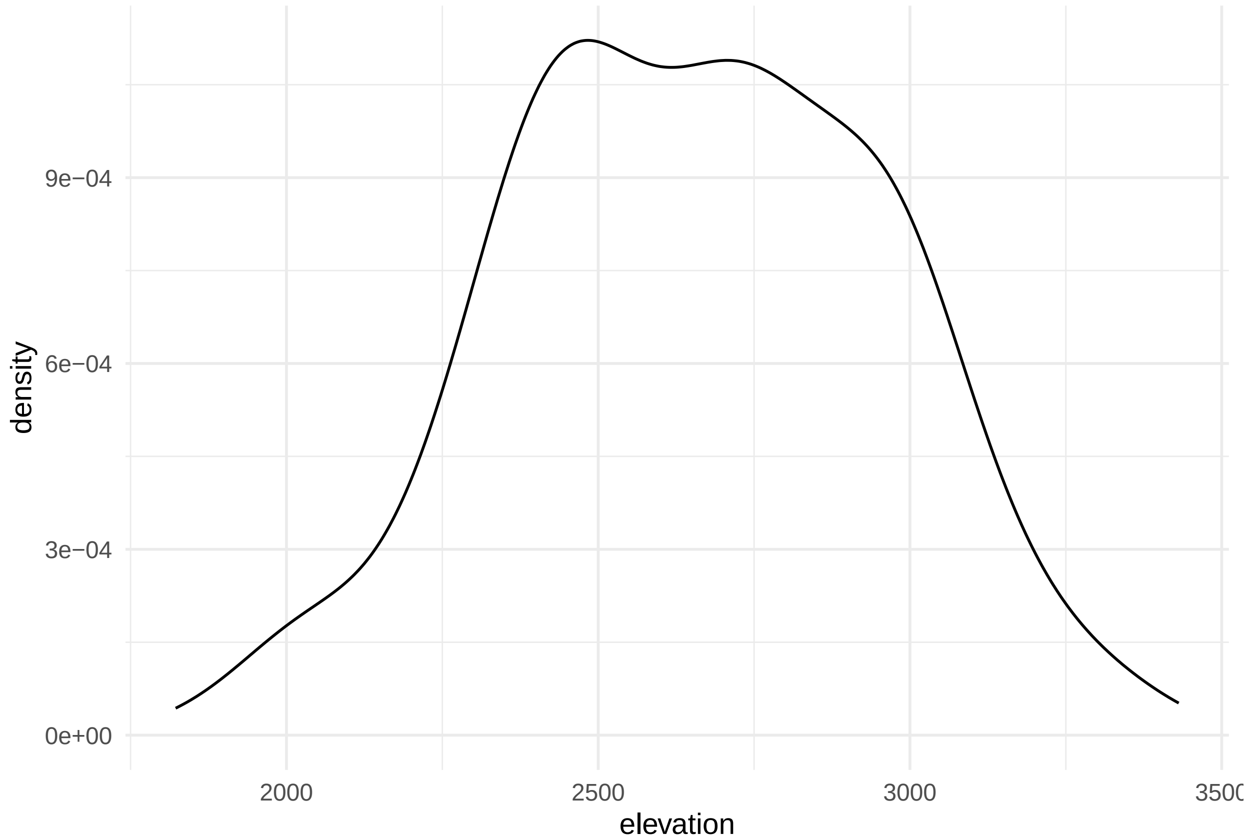
```
## Warning: Removed 206 rows containing non-finite outside the scale range
## (‘stat_density()‘).
```

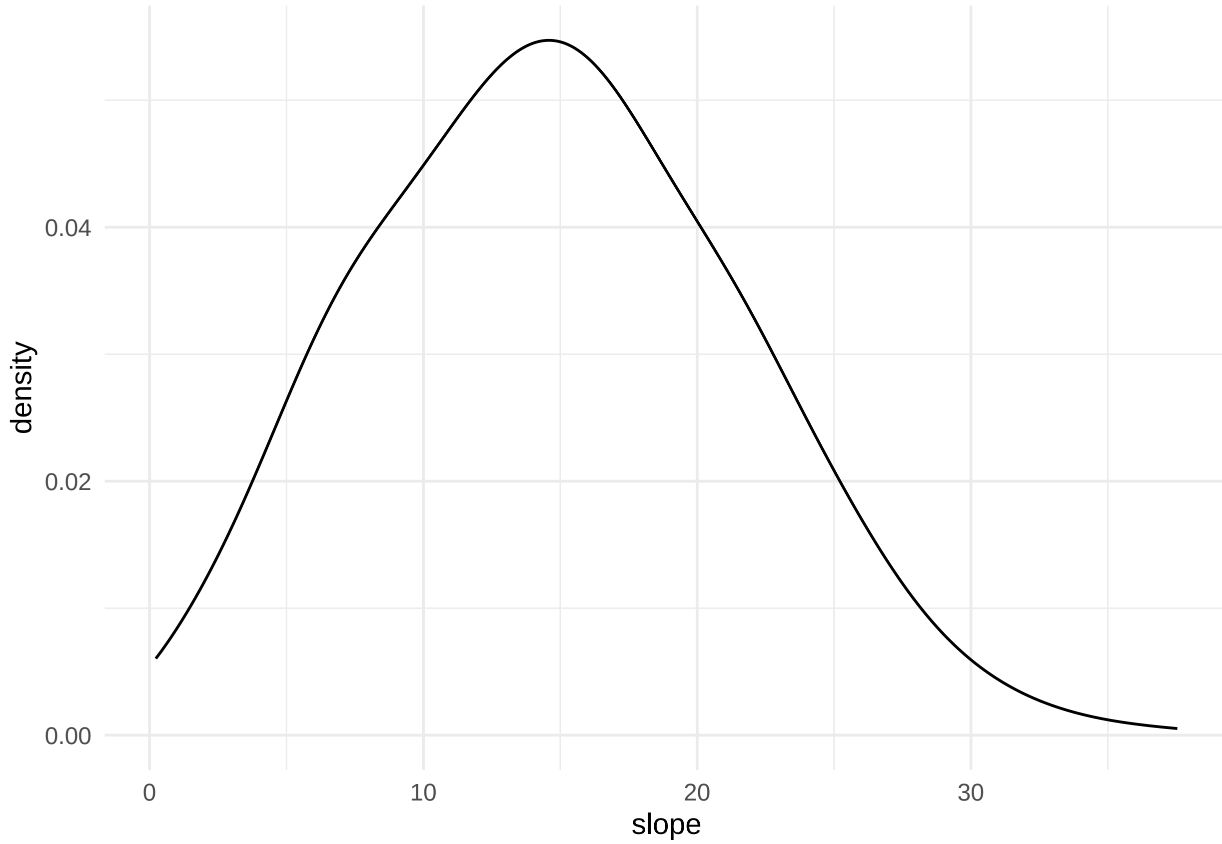


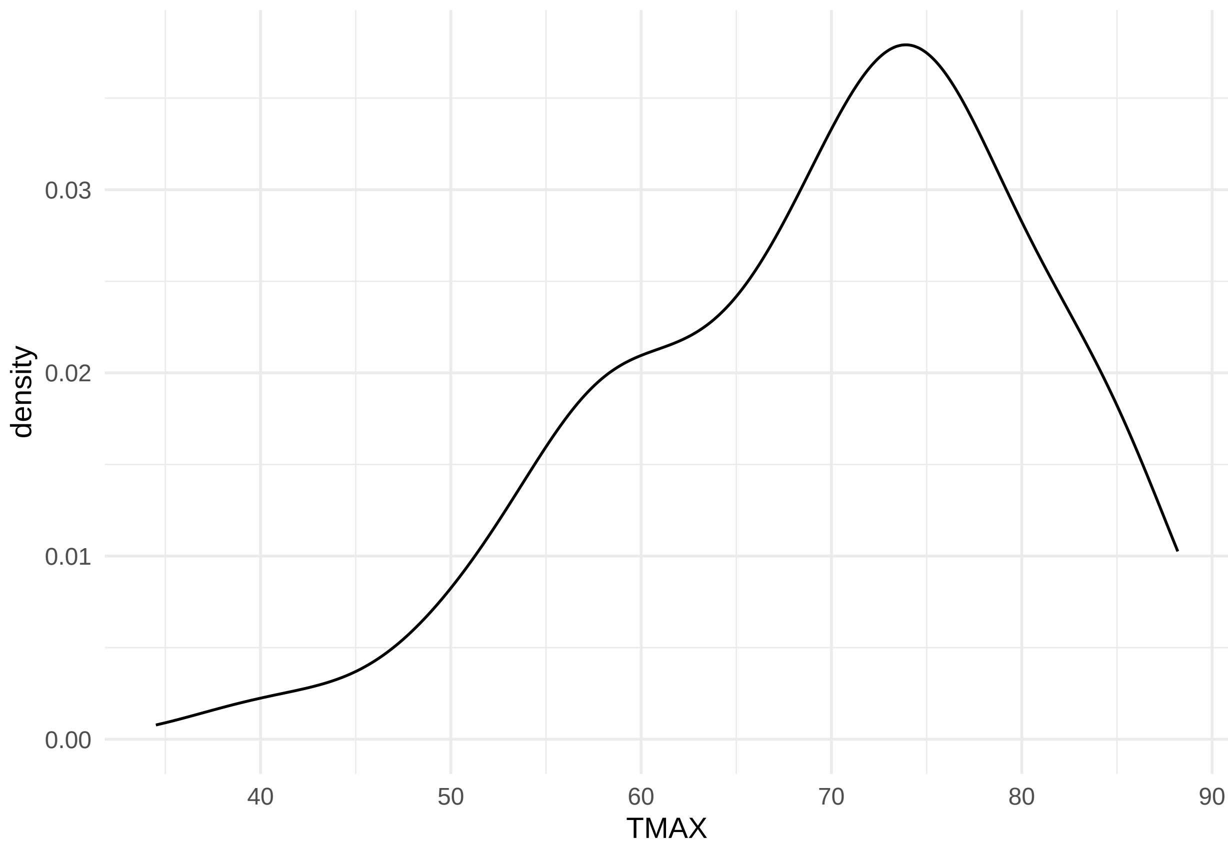

```
## Warning: Removed 206 rows containing non-finite outside the scale range
## (‘stat_density()‘).
```

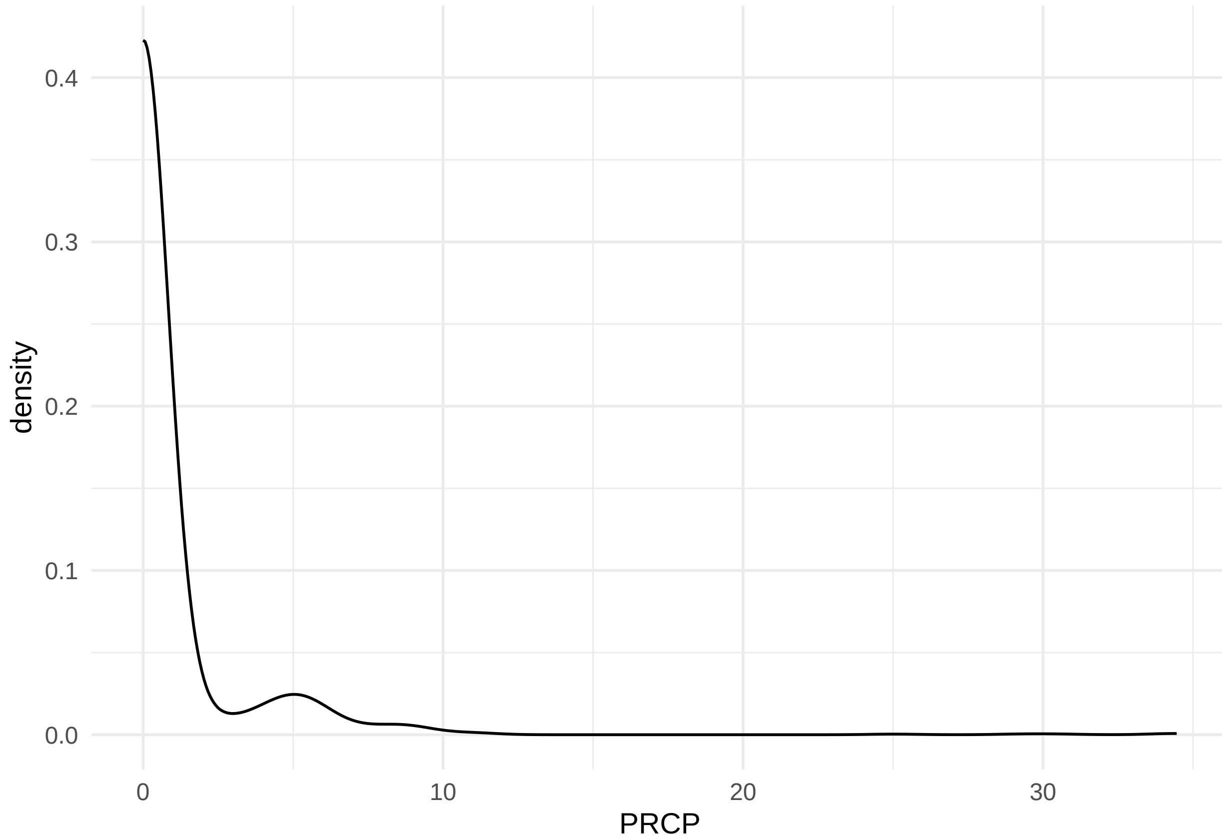


density
0.0
0.2
0.4
0.6
−1.0
−0.5
0.0
0.5
1.0
eastness

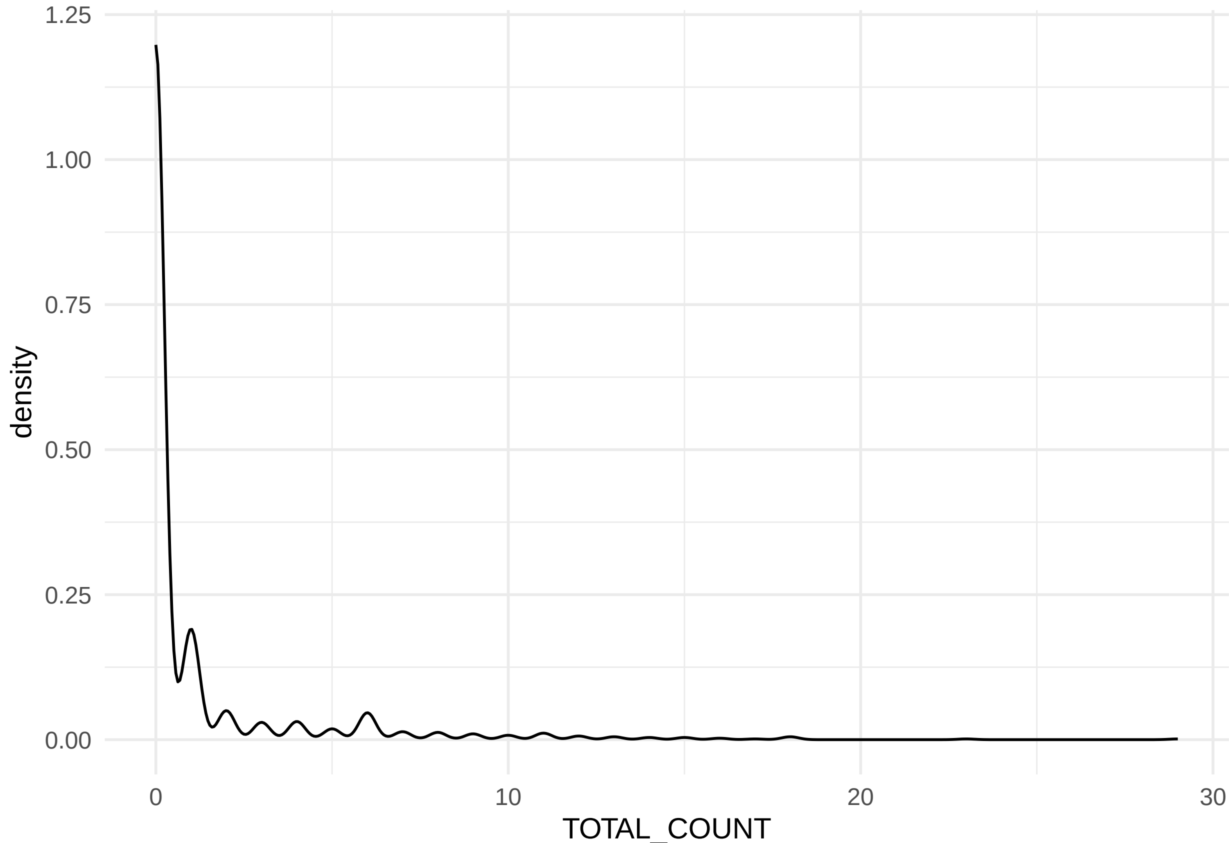

density
9e−04
6e−04
3e−04
0e+00
2000
2500
3000
3500
elevation


density
0.00
0.02
0.04
0
10
20
30
slope

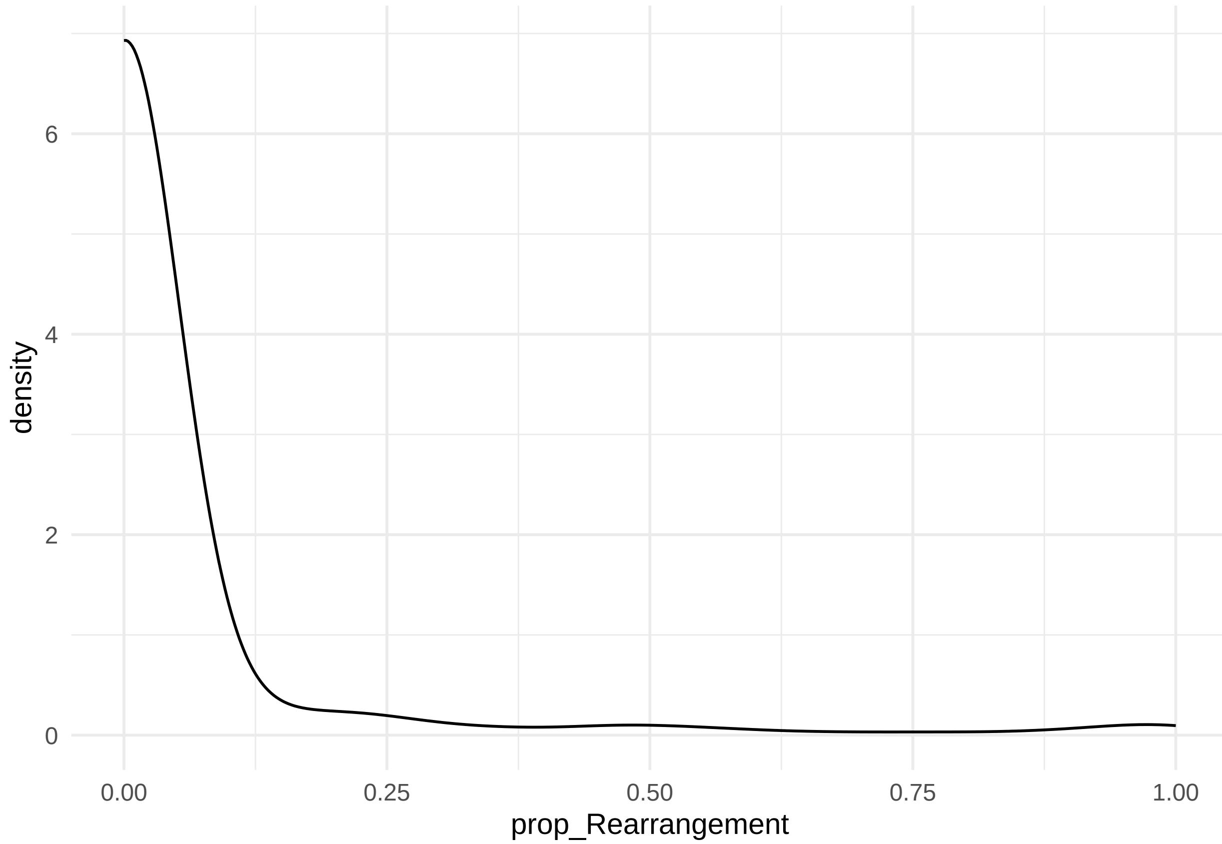

0.03
0.02
0.01
0.00
density
40
50
60
70
80
90
TMAX



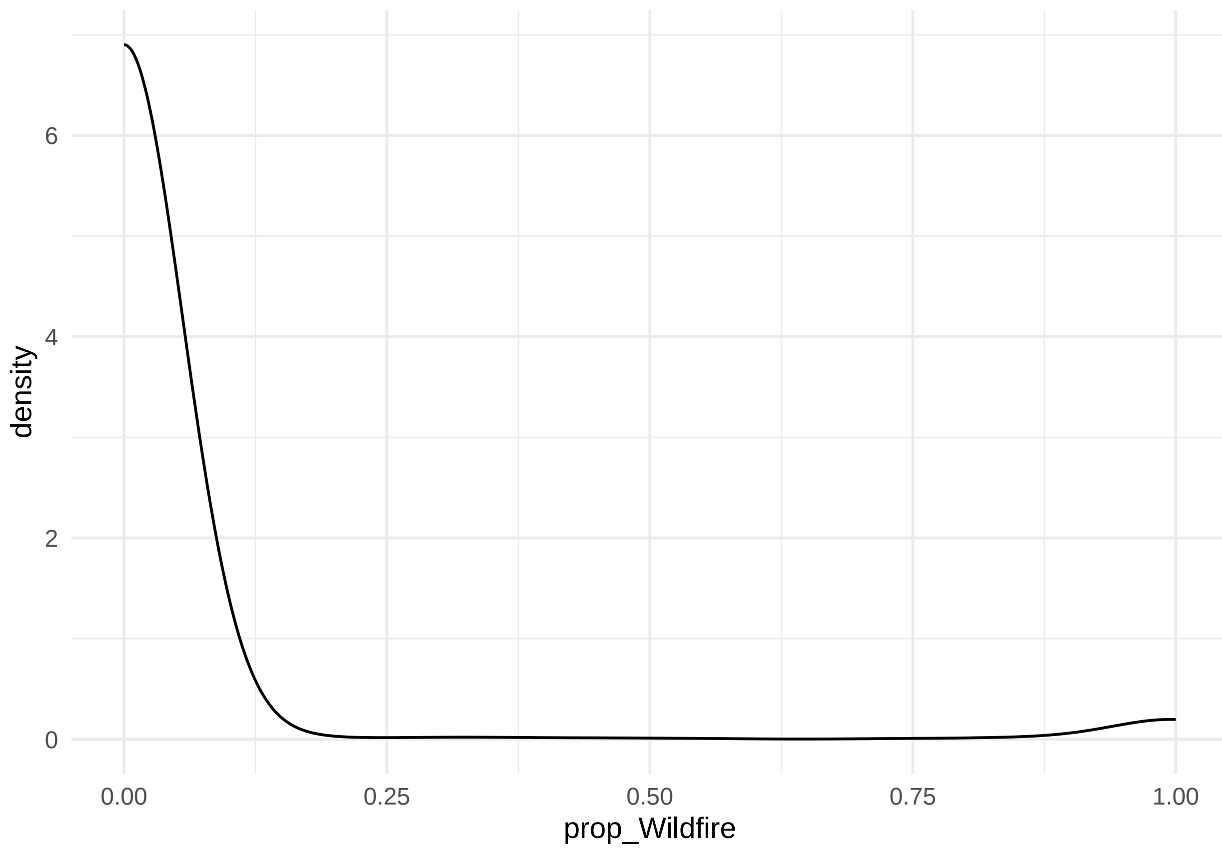
density
0.4
0.3
0.2
0.1
0.0
0
10
20
30
PRCP

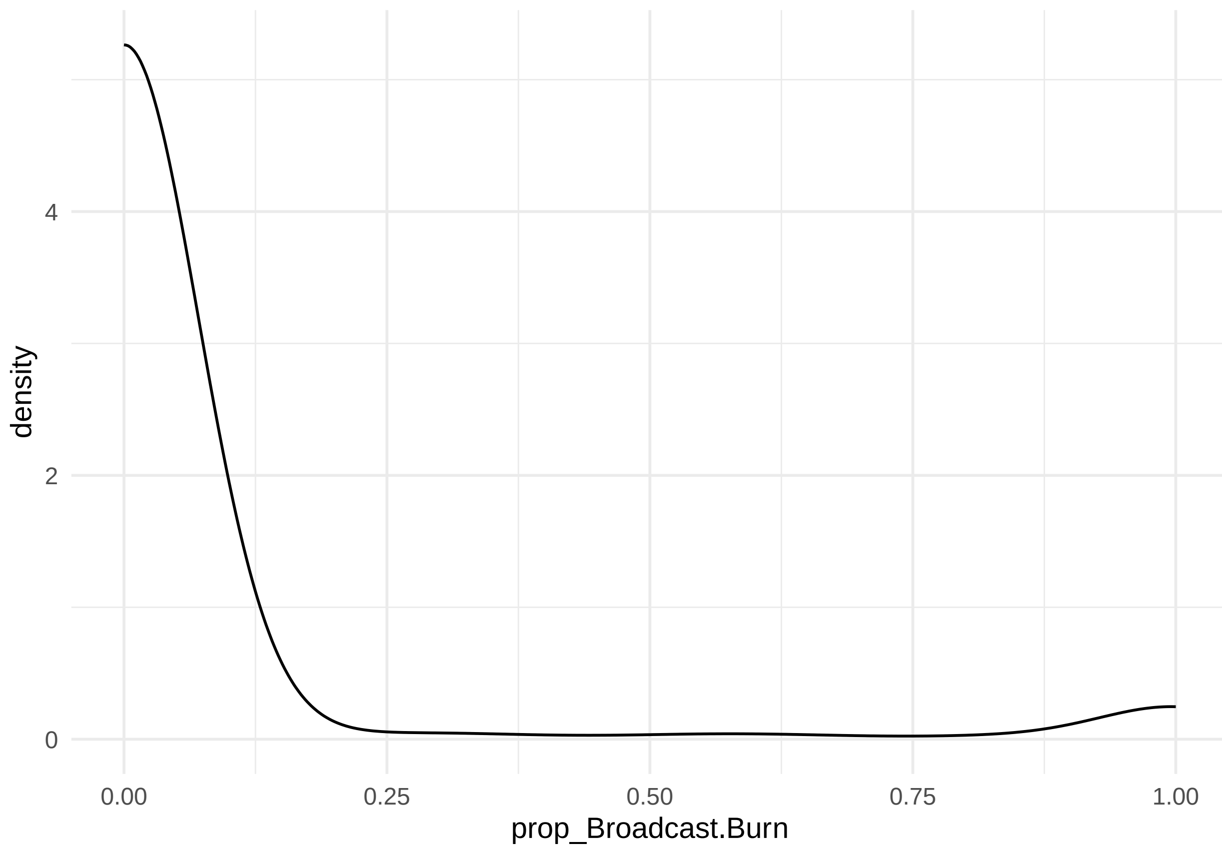


1.25
1.00
0.75
0.50
0.25
0.00
density
0
10
20
30
TOTAL_COUNT

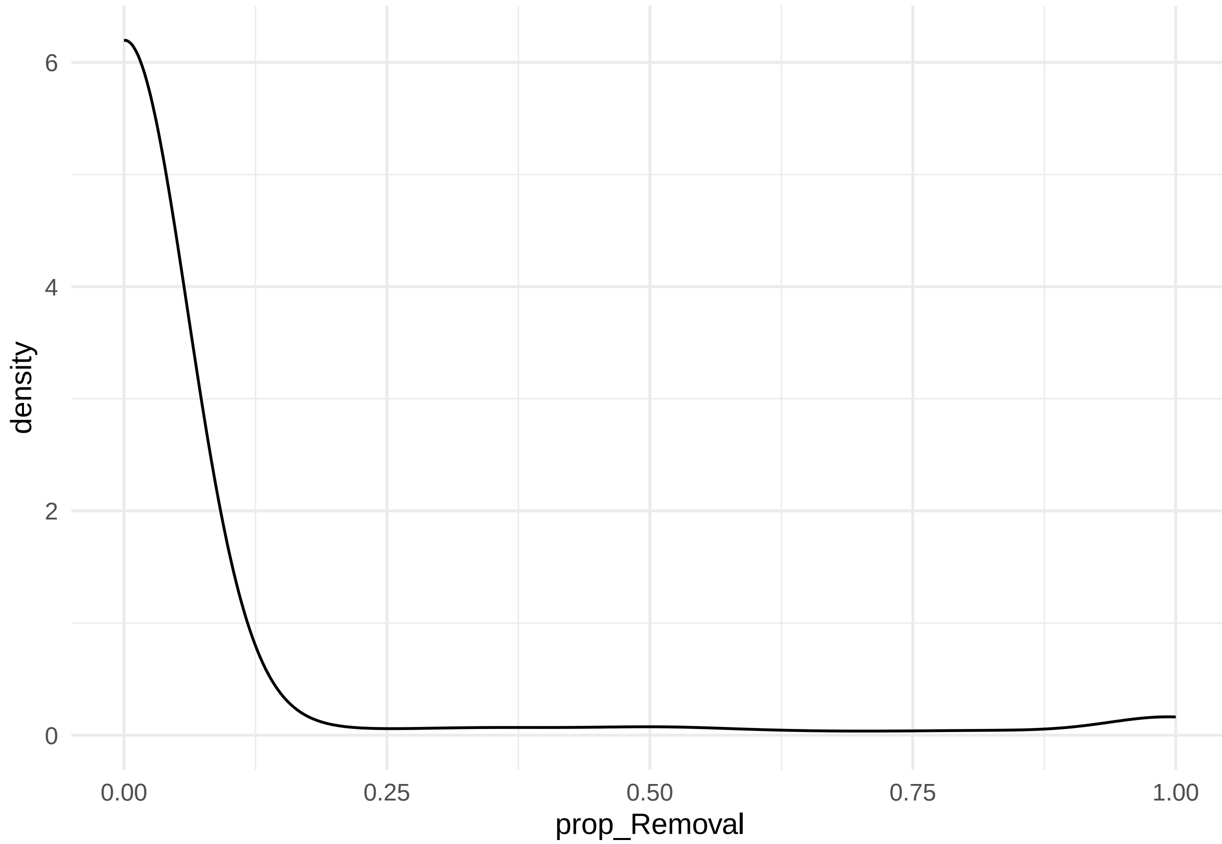


density
6
4
2
0
0.00
0.25
0.50
0.75
1.00
prop_Rearrangement

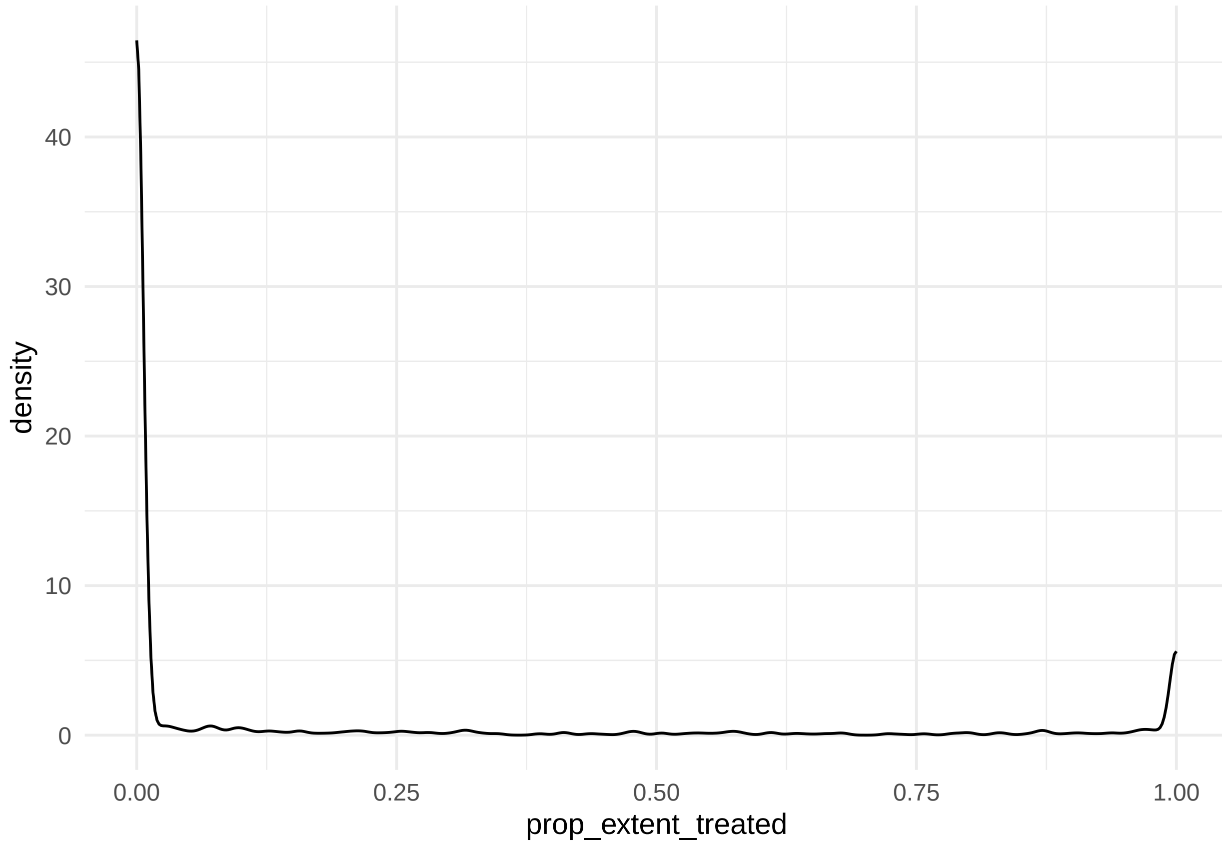

6
4
2
0
density
0.00
0.25
0.50
0.75
1.00
prop_Wildfire

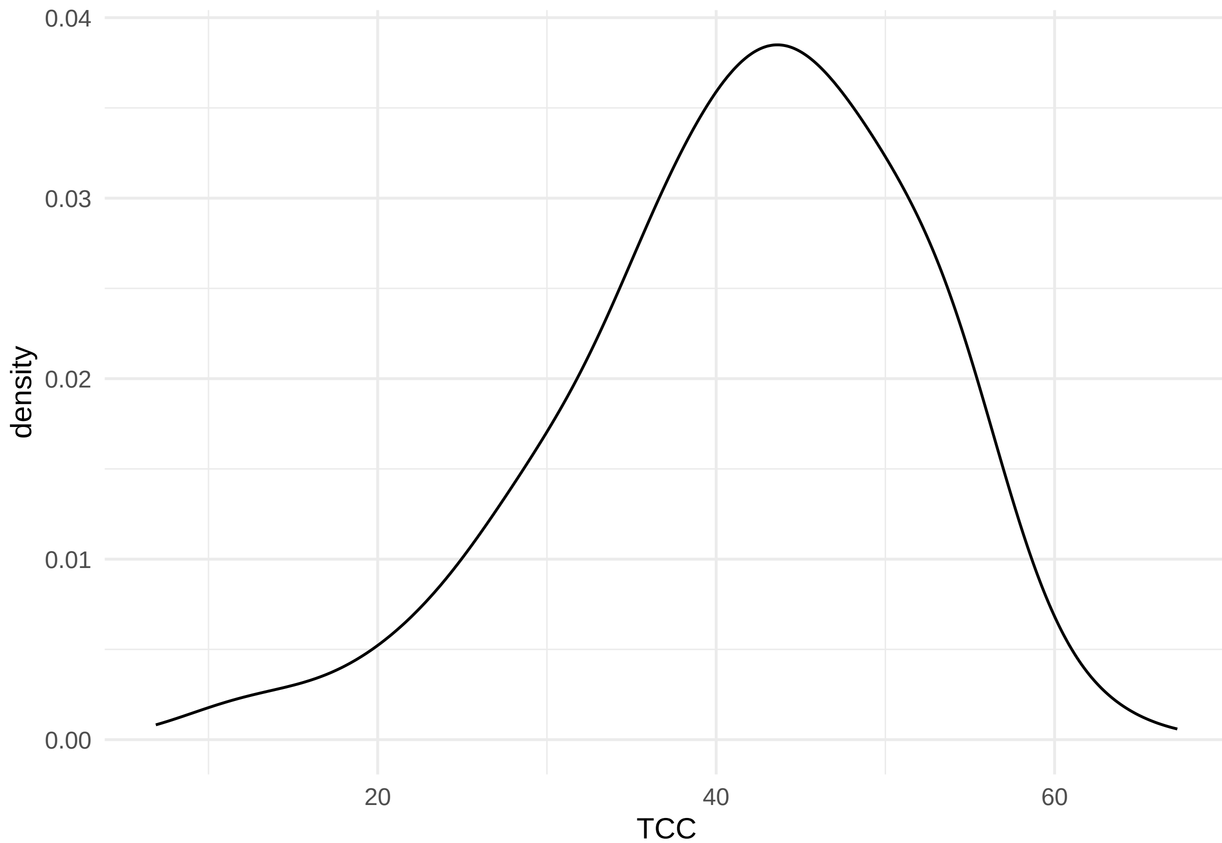


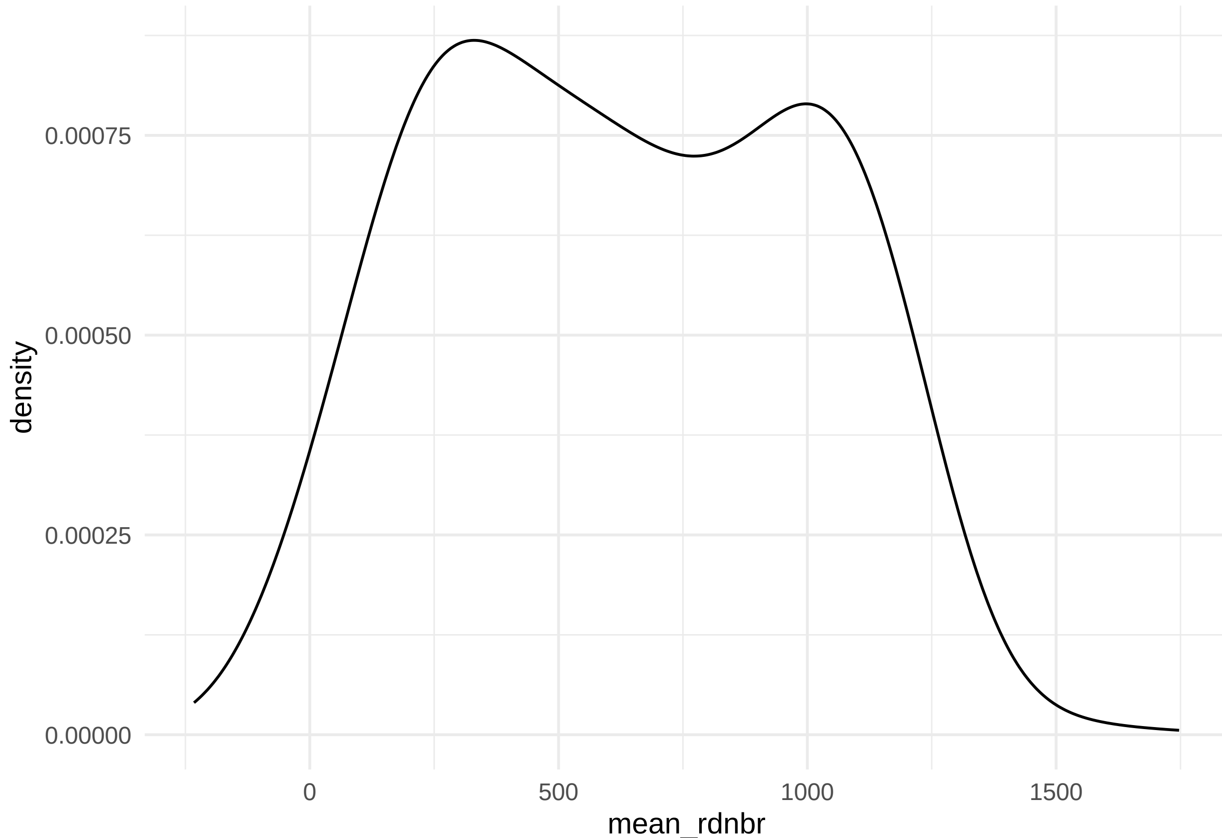
density
0
2
4
0.00
0.25
0.50
0.75
1.00
prop_Broadcast.Burn



6
4
2
0
density
0.00
0.25
0.50
0.75
1.00
prop_Removal


density
40
30
20
10
0
0.00
0.25
0.50
0.75
1.00
prop_extent_treated


0.04
0.03
0.02
0.01
0.00
density
20
40
60
TCC


0.00075
0.00050
0.00025
0.00000
density
0
500
1000
1500
mean_rdnbr